

From Spikes to Symbols

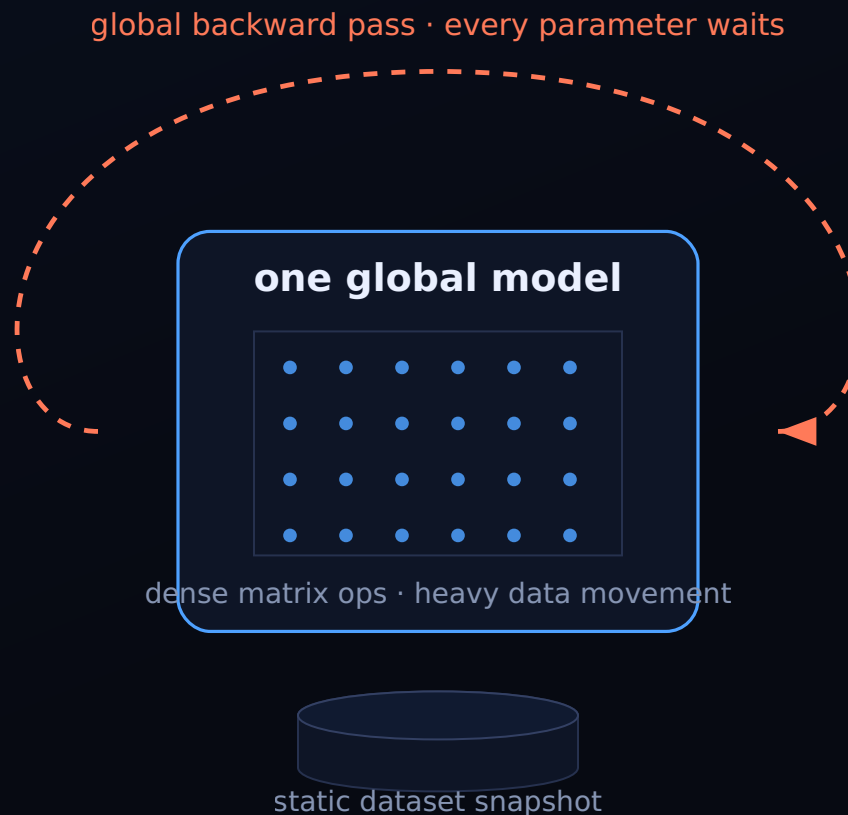
Growing intelligence from many local circuits that learn continuously — instead of compressing it into one globally trained model.



The limits of globally trained intelligence

LLMs are extraordinary. But their power rests on structural commitments worth questioning.

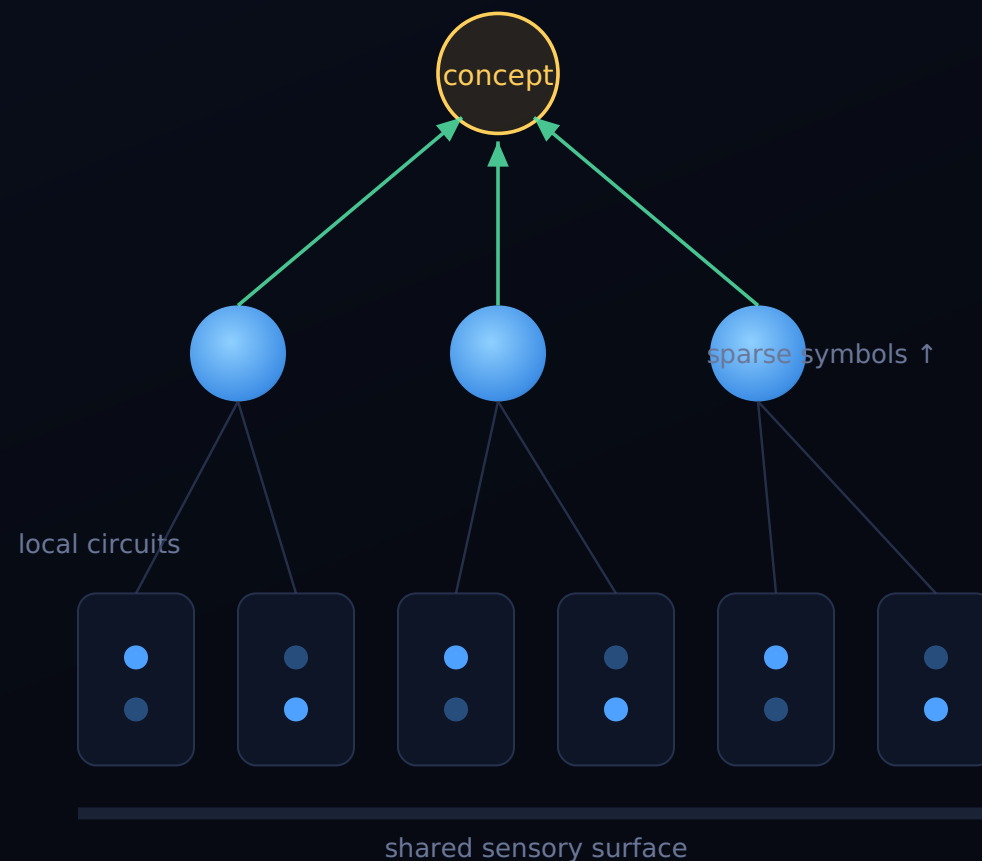
- Knowledge is learned by one **global optimization** over a large, mostly static dataset.
- Updating it can mean **retraining** or bolt-on external memory.
- Representations are powerful but **distributed and hard to inspect**.
- The model is generally asked to emit **another token even when its evidence is weak**.



Intelligence from local circuits

Not one model that sees everything at once — many small circuits that each learn from what is near them, and pass sparse symbols upward.

- **Local receptive fields** and event-driven spikes
- **Local, continuous learning** — no global error signal
- **Excitation and inhibition** create stable specialists
- **Hierarchy** composes symbols into concepts



Biology computes locally

Nervous systems don't start with a global model reading every value at once. They start with many local circuits that turn nearby events into features.

membrane charge accumulates

neurons spike

refractory pacing

excitation reinforces

inhibition competes

synapses learn locally

specialists emerge

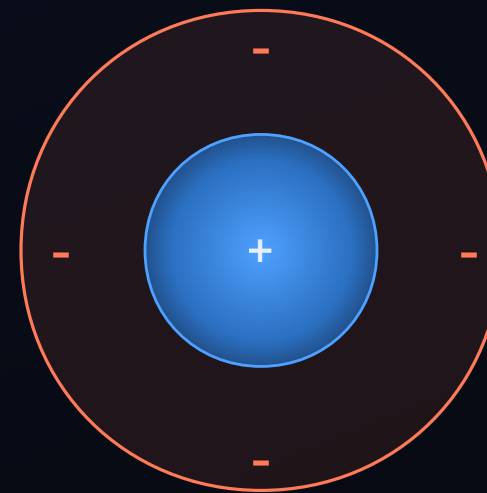
a neuron integrates, then fires



The retina: center-ON, surround-OFF

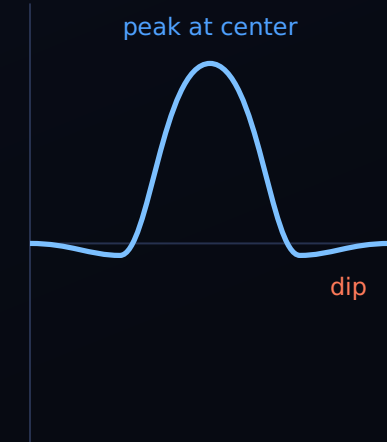
A ganglion cell doesn't report raw light. It compares a small center against its surround — signalling local **contrast**, not a copy of every value.

- Center excites; surround inhibits
- Output encodes **structure**, not brightness
- **Lateral inhibition** separates competing interpretations



center-ON / surround-OFF field

response



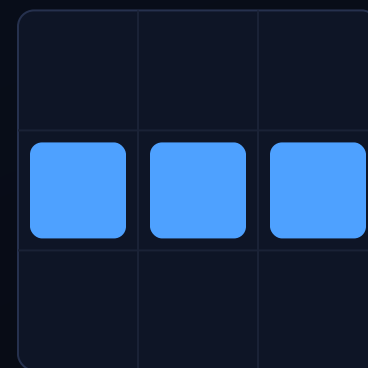
position across field →

A 3×3 window learns oriented lines

A center-crossing 3×3 patch is the smallest neighborhood where a cell can learn relationships between a center and its surroundings.

- Four primitives share the center pixel
- Local cells become **orientation-sensitive**: horizontal, vertical, two diagonals
- These features become inputs to the next layer

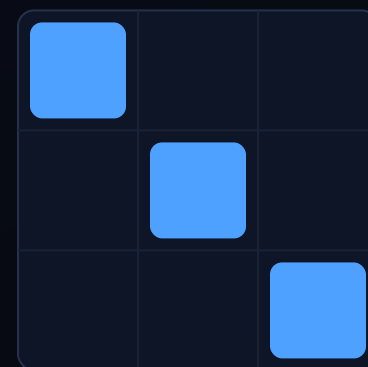
The beginning of a retinal-to-cortical hierarchy — built from repeated local windows.



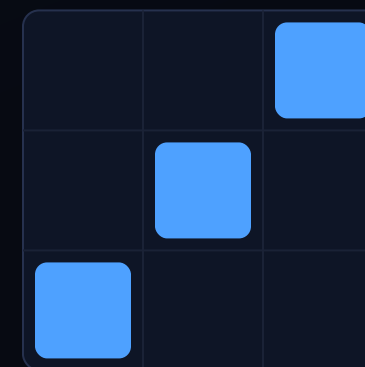
horizontal



vertical



diagonal \



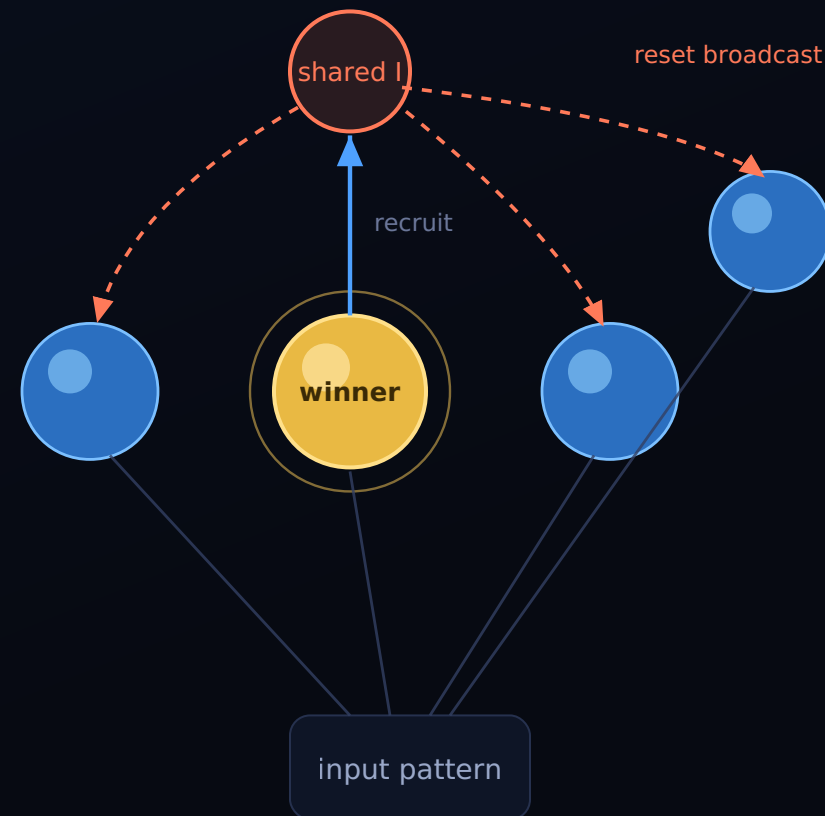
diagonal /

Competition creates specialist neurons

Many neurons compete to explain an input. A shared inhibitory neuron broadcasts a reset, so the best-matched unit wins repeatedly — and specializes.

- Winner fires and strengthens its match
- A shared **inhibitory neuron** resets the pool
- Losing capacity is **reallocated**, recruiting new specialists

■ excitatory (E) ■ inhibitory (I)

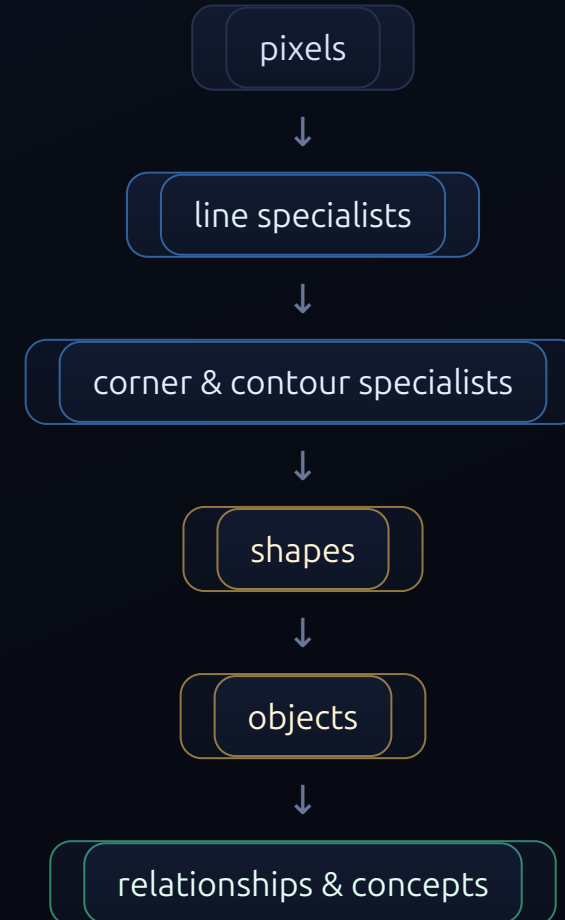


A stable specialist becomes a symbol

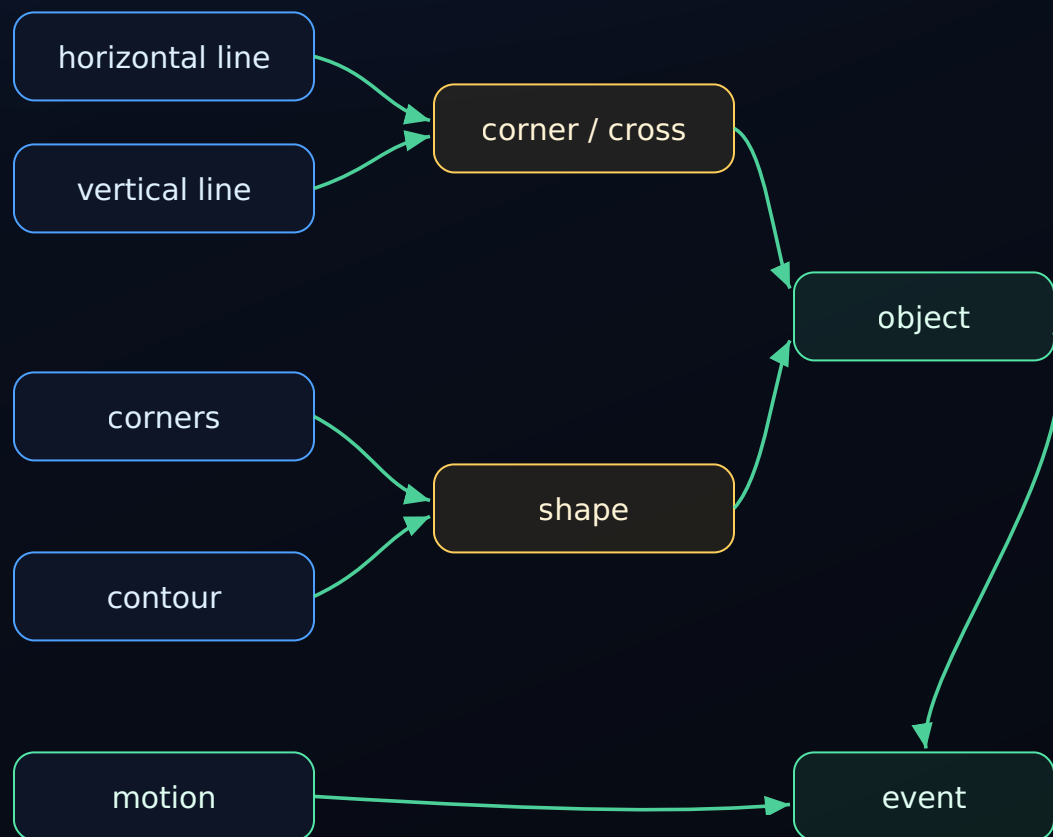
Once a neuron reliably and selectively answers to a recurring structure, its spike *is* that structure — a compact, reusable token.

Symbol ownership: a specific internal assembly responds reliably and selectively to the same meaningful structure.

- A persistent internal identity
- Recognition is **separate from language**
- A unit higher layers can compose



Hierarchies make symbols reusable



A higher layer never relearns raw pixels. It learns **combinations of lower-level symbols**.

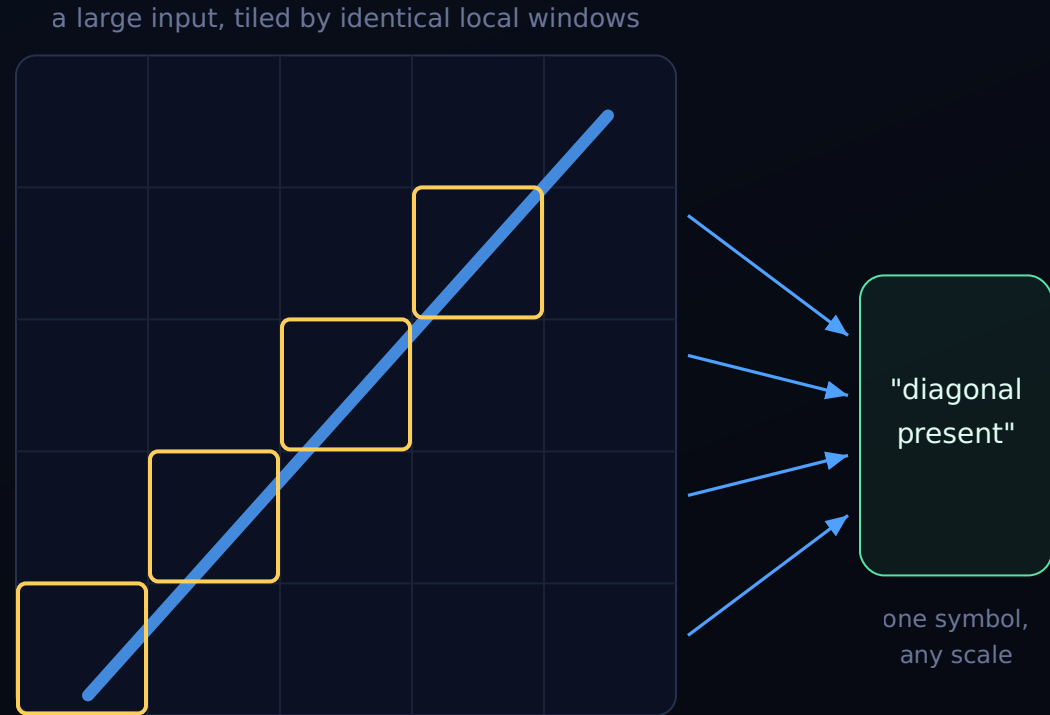
- **Reuse:** one primitive joins many larger concepts
- **Combinatorial generalization:** known pieces form unseen combinations
- **Modularity:** add concepts without redesigning the network
- **Interpretability:** trace a concept back through its parts

Size-agnostic learning through local features

A line is a line — in a 3×3 patch, inside a large image, at another position, as part of a letter or an object.

- Scale by **repeating local circuits** across space
- Each emits sparse symbols upward
- Higher layers consume **recognized features**, not every raw pixel

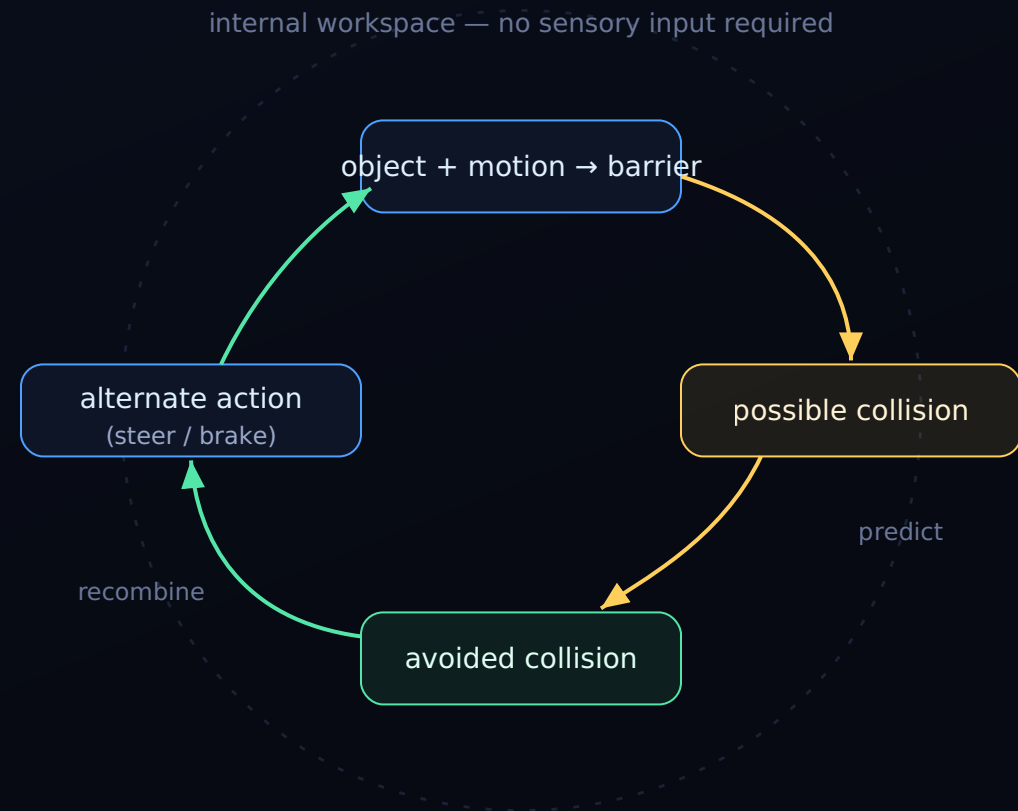
Larger inputs need more receptive fields and routing
— **not a different learning rule.**



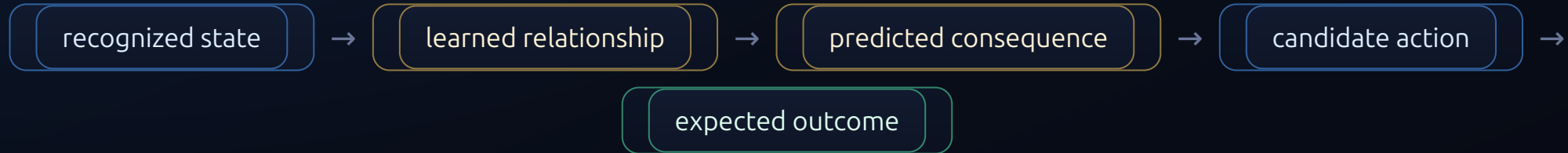
Composition becomes imagination

A stable symbol no longer needs the original sensory input to activate. Higher circuits can reactivate and recombine symbols into **candidate situations**.

- Construct configurations never directly observed
- Compare possible future states
- Test alternative actions internally
- Learn from prediction–observation mismatch



Symbols become reasoning chains



THIS ARCHITECTURE

- Each step is an **owned symbol** or assembly you can inspect
- A known transition is distinguishable from a missing one
- Failed predictions update the **responsible local relationship**
- Useful chains become reusable higher-level symbols

AN LLM

- Reasoning is expressed as a **sequence of tokens** from distributed representations
- Here, reasoning is a sequence of **owned symbols and learned transitions**
- Language can later *describe* the chain — but language is not the chain itself

Symbols are grown, not authored

Expert systems and neuro-symbolic pipelines **import** their symbols and rules. Here both are **discovered** by local learning and grounded in perception.

	Expert systems & ontologies	Neuro-symbolic (NeSy)	This architecture
Symbols	Hand-authored vocabulary	Predefined predicate set	Emergent — a neuron earns symbol status by winning competition
Meaning	Ungrounded tokens	Perception grounded, symbols still given	Grounded by construction — the symbol is its detector
Reasoning	Separate inference engine over a rule base	Neural front-end + a distinct logic module	Learned transitions in the same substrate
Learning	None / manual KB updates	Grounding learned; trained globally	Continuous, local, gradient-free

The goal: the virtues of the symbolic level — inspectable, compositional, uncertainty-aware — **without authoring an ontology or grounding tokens after the fact.**

Knowing, uncertainty, and the unknown

Knowledge should require a **stable, strong internal representation** — not merely producing an output.

Uncertainty emerges from observable dynamics: response strength, competitive margin, stability over time, and whether a reasoning chain completes.



"I recognize this."

One stable specialist with a strong competitive margin.



"I'm uncertain which."

Several nearly equal specialists — no clear winner.



"Known parts, new whole."

Recognized components in an unfamiliar arrangement.



"I have no representation."

No specialist reaches the required response — flag novelty, stop.

Continuous learning changes the compute equation

Learning and inference run through the **same local circuit**. A new event updates only the synapses it touches — the system keeps operating.

- No network-wide backward pass, no global loss
- No stored activation graph for backprop
- **Sparse updates** tied to active events
- No replay of the full historical dataset per update

Rebuild a model of the world



frozen between costly global rebuilds

Adapt while operating in the world



every event is incorporated immediately

learning cost $\propto$ active events

less data movement · one substrate for inference & learning

Less dependence on static datasets — and more visible bias

It doesn't remove the need for data; it changes data's role — from one giant frozen snapshot toward **ongoing experience**.

- Immediate incorporation of new information
- Adaptation to local, changing conditions
- Fewer examples for concepts built from known parts

Bias isn't automatically solved — but it becomes **measurable representational coverage** instead of diffuse opacity.

Which concepts own a specialist?



- Green owned specialist
- Brown colliding / undifferentiated
- Red over-represented / dominant
- Dashed no representation (gap)

gaps are inspectable, not hidden across billions of parameters

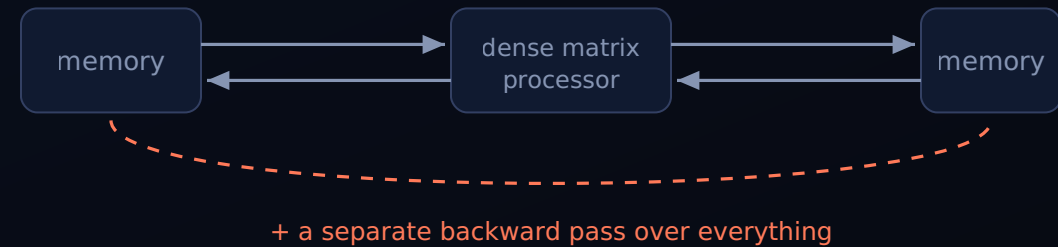
Scaling onto neuromorphic hardware

The theory maps naturally to spiking chips: event-driven, sparse, with memory and compute in the same place.

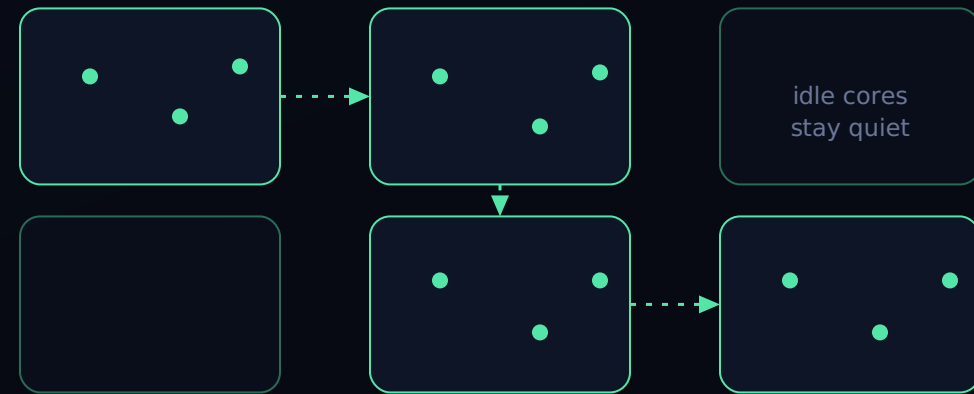
- Local state · local synapses · spike routing
- Learning directly on the device
- Low activity when the world is unchanged
- Unrelated regions run **independently** — no global gradient to wait on

Especially valuable for always-on robotics, edge devices, and adaptive sensors under limited power.

Conventional accelerator



Neuromorphic system — replicated cores, learning at the point of activity



What is predicted vs. what is measured

ARCHITECTURAL PREDICTIONS

The composition, imagination, reasoning, and scaling described here are what the architecture is **designed to enable**. They follow from its principles — local receptive fields, owned symbols, local learning — and describe the **long-term hypothesis** this research pursues.

MEASURED TODAY

The working system is a small, from-scratch spiking network learning 3×3 line primitives with local rules and no gradients. Concrete **speed, energy, and memory advantages on neuromorphic hardware are future validation** — stated as expected sources of gain, not benchmarked numbers.

The gap between these two columns is the research program — not a claim already banked.

THE THESIS, IN ONE SENTENCE

Instead of training one enormous model to approximate everything it has seen, we can build intelligence from **local systems that learn continuously**, form **explicit symbols**, compose new possibilities, trace their own reasoning, and **recognize the boundaries of their knowledge**.

From Spikes to Symbols · a theoretical architecture

Sources & grounding

Repository documents (this work)

- **README.md** — architecture, learning rule, task
- **Current_Implementation_Methodology_Equations.md** — the equations behind every mechanism
- **Architecture_Decisions_Report.md** — decision log, locality & biology assessment
- **Inhibitory_Off_Weight_Recruitment_Spec.md** — competitive reset & recruitment
- **PaulVersion.md** — the "1 vs Z" paradigm spec
- **docs/DASHBOARD.md** — live simulation dashboard
- Core code: **neuron_flexible.py**, **cortical_column_flexible.py**, **backend/simulation.py**, **snn/rules/**

Scientific motifs referenced

- Leaky integrate-and-fire neurons — Gerstner & Kistler
- Center-surround retinal receptive fields; lateral inhibition — Kuffler; Hartline
- Cortical winner-take-all microcircuits — Douglas & Martin; Amari
- Spike-timing / eligibility-trace plasticity — Bi & Poo
- Inhibitory (iSTDP) plasticity for E/I balance
- Homeostatic synaptic scaling — Turrigiano
- Sparse, highly selective coding — Quiroga et al.

Motifs are cited as biological inspiration, not as a claim of faithful simulation of any specific cell type. Full speaker notes accompany the deck in **SPEAKER_NOTES.md**.